



Some thesis title

by

Mostert

*Thesis presented in partial fulfilment of the requirements for  
the degree of Doctor of Philosophy (Electronic Engineering)  
in the Faculty of Engineering at Stellenbosch University*

Supervisor: Prof. Name Surname

2026

*Someone.*

---

# ACKNOWLEDGEMENTS

---

\* Supervisor X.

\* Funding Y.

“ *A very long quote*  
*from someone very important.* ”

– Mr M

Thank you to Leanne for helping to prepare this template.

By submitting this thesis electronically, I declare that the entirety of the work contained therein is my own, original work, that I am the sole author thereof (save to the extent explicitly otherwise stated), that reproduction and publication thereof by Stellenbosch University will not infringe any third party rights and that I have not previously in its entirety or in part submitted it for obtaining any qualification.

Copyright © 2026 Stellenbosch University  
All rights reserved.

---

# ABSTRACT

---

---

# UITTREKSEL

---

---

# TABLE OF CONTENTS

---

---

# LIST OF FIGURES

---

---

# LIST OF TABLES

---

---

# NOMENCLATURE

---

## ACRONYMS AND ABBREVIATIONS

**MFT** My Funky Thesis



# INTRODUCTION

---

## 1.1 BACKGROUND

Municipal wastewater treatment works (WWTW) are made up of several process units that work together in sequence. Larger plants usually rely on the activated sludge process, which uses biological reactors full of suspended solids (activated sludge) plus secondary settling tanks called clarifiers.

Inside the biological reactors, treatment happens in unaerated and aerobic zones. The aerobic zones need mechanical aeration to push oxygen into the water so bacteria can break down waste. There are several ways to do this, but Fine Bubble Diffused Aeration (FBDA) is one of the most common. Once the water has been aerated, it moves into circular clarifiers, which separate clear effluent from the sludge. Gravity pulls the sludge down into a central hopper, where it gets sent back to the bioreactor to treat more incoming wastewater.

Interestingly, someone who knows what to look for can tell a lot about how these systems are performing just from aerial or satellite photos. A healthy FBDA unit churns the surface into a "leopard print" pattern, while a struggling one looks noticeably smoother. Clarifiers tell a similar story: when they're working well, they separate out the solids and leave behind a clear liquid that looks black from above. When they're not working, solids escape into the effluent and the surface turns green, cloudy, or uneven.

## 1.2 PROBLEM STATEMENT

Right now, checking on how well WWTW units are running mostly comes down to on-site sensors, physical inspections, or someone experienced reading aerial photos by eye. That's slow, and it doesn't scale well once you're trying to keep tabs on multiple facilities at once. The question this project asks is simple: can a computer vision system pick up on

the same visual cues a trained person would (the aerator's "leopard print," the clarifier's colour) and use them to judge performance? This project tests whether historical satellite imagery can be used to automatically monitor how well FBDA aerators and clarifiers are functioning.

## 1.3 OBJECTIVES

To answer that question, the project sets out to do three things:

1. Prepare the data and extract features by working with a set of labelled Google Earth satellite images taken on different dates to capture the visual states of WWTW components.
2. Classify the states using a CNN by training a convolutional neural network to sort Aerobic Zones into three categories (Functional, Suboptimal, Dysfunctional) and Clarifiers into four categories (Functional, Dysfunctional, Scum, Empty), based on how their surfaces look.
3. Test whether it actually works .....

## 1.4 SUMMARY OF WORK

## 1.5 SCOPE

Included: Fine Bubble Diffused Aeration (FBDA) in aerobic zones, and secondary clarifiers. The project only looks at the Atlantis, Cape Flats, and Waterval WWTW facilities.

Excluded: Surface-mounted aerators and anaerobic zones aren't covered. There's no live satellite feed, no physical water sampling, and no hardware devices deployed on site — everything is based on the historical Google Earth imagery already available.

## 1.6 ROADMAP

- Chapter 2 (Literature Review): Looks at existing monitoring solutions for wastewater management, remote sensing tech, and the computer vision concepts this project builds on.
- Chapter 3 (System Methods): Covers the system's architecture, how the dataset is structured, and the metrics used to judge the model's performance.
- Chapter 4 (Detailed Design): Gets into the technical details of the image processing pipeline and the classification logic.

- Chapter 5 (Results): Presents the data, the performance numbers, and what they say about how accurately the system classifies each unit.
- Chapter 6 (Conclusion): Wraps up the findings, checks how well the objectives were met, and suggests where future versions of this kind of monitoring could go.

# LITERATURE

---

## 2.1 RELATED WORK

# SYSTEM METHODS

---

## 3.1 DATASET

The dataset used in this project consists of satellite imagery sourced from Google Earth, covering three WWTW facilities: Atlantis, Cape Flats, and Waterval. Each image is a snapshot of a facility's biological reactors at a particular point in time, with filenames encoding the capture date in a `month_year` format (e.g. `03_2024` for March 2024). Where more than one image was available for the same month, this is indicated with a number in brackets, e.g. `Waterval 07_2016 (2)`.

Each biological reactor in the dataset consists of an aerobic zone and one or more clarifiers, and these are the two components the project focuses on classifying.

### 3.1.1 ANNOTATIONS

Images were manually annotated using the VGG Image Annotator (VIA2) tool. Aerobic zones were labelled using rectangular bounding boxes, while clarifiers were labelled using circular regions, reflecting the physical shape of each structure. Each annotation carries a region attribute indicating the visible state of that component at the time the image was captured.

Aerobic zones were labelled into one of three categories, based on how much of the surface exhibits the characteristic "leopard print" aeration pattern:

- \* **Functional:** the surface shows the leopard print pattern.
- \* **Suboptimal:** a third or more of the surface shows no leopard print, or the state is unclear.
- \* **Dysfunctional:** two thirds or more of the surface shows no leopard print, or is clearly dysfunctional.

Clarifiers were labelled into one of four categories, based on surface colour and visibility:

- \* **Functional**: surface appears uniformly black.
- \* **Dysfunctional**: surface appears green.
- \* **Scum**: surface is not uniformly black.
- \* **Empty**: shadows are visible inside the tank.

Annotations are stored as VIA2 project files in JSON format. Each entry records the image filename, file size, and a list of regions, where each region specifies its shape (rectangle or circle), pixel coordinates, and the associated class label. An example entry for an aerobic zone annotation is shown below:

```
"Atlantis 01_2019.jpg3469910": {
  "filename": "Atlantis 01_2019.jpg",
  "size": 3469910,
  "regions": [
    {
      "shape_attributes": {
        "name": "rect",
        "x": 2707, "y": 931,
        "width": 2839, "height": 1381
      },
      "region_attributes": { "aerobic zone": "Dysfunctional" }
    }
  ],
  "file_attributes": {}
}
```

### 3.1.2 FACILITY STRUCTURE

The three facilities differ slightly in how their data is organised, reflecting differences in plant layout.

**Atlantis** consists of a single biological reactor, so its 48 images are stored in one folder with no subdivisions. Aerobic zone and clarifier labels are each stored in a single JSON file. An unlabelled overview image of the plant is also included for reference.

**Cape Flats** consists of four separate reactor units (CF1–CF4), each with its own image folder and containing between 92 and 100 images. Each unit has its own aerobic zone and clarifier label file, giving eight JSON files in total. An overview image of the

whole plant is included, along with a version showing the unit numbering, to make it clear which physical unit each folder corresponds to.

**Waterval** consists of two reactor units, referred to as Top and Bottom, with 37 and 36 images respectively. As with Cape Flats, each unit has its own aerobic zone and clarifier label file. An unlabelled overview of the full plant is also included.

Table ?? summarises the structure of the dataset across the three facilities.

**Table 3.1:** Summary of the dataset by facility

| Facility   | Units | Images             | Label files |
|------------|-------|--------------------|-------------|
| Atlantis   | 1     | 48                 | 2           |
| Cape Flats | 4     | 382 (96+94+92+100) | 8           |
| Waterval   | 2     | 73 (37+36)         | 4           |

This structural difference between facilities — a single unit at Atlantis, four at Cape Flats, and two at Waterval — is factored into the data pipeline described in Section ??, since each facility’s images and labels need to be parsed slightly differently before they can be fed into the CNN.

## SUMMARY AND CONCLUSIONS

---

---

# REFERENCES

---

- [1] Stellenbosch University, “Formatting your thesis,” 2024, <http://bit.ly/SUThesisFormat>.
- [2] D. Els, “Stellenbosch university thesis bundle,” 2024, <https://ctan.org/pkg/stellenbosch-2>.

## APPENDIX A

---

APPENDIX TITLE GOES HERE

---